
Corrigé — Exercice 16

1) $i\bar{z} - 4 = 0 \Rightarrow \bar{z} = -4i \Rightarrow z = 4i$. $z^2 - 4z + 16 = 0 : \Delta = -48 = (4\sqrt{3}i)^2 \Rightarrow z = 2 \pm 2\sqrt{3}i$.

$$S = \{4i, 2 + 2\sqrt{3}i, 2 - 2\sqrt{3}i\}$$

2) $z_A = 4, z_B = 4i, z_C = 2 - 2\sqrt{3}i, z_D = -iz_C = -2\sqrt{3} - 2i$.

2a) $|z_A| = |z_B| = |z_C| = |z_D| = 4 \Rightarrow A, B, C, D \in \mathcal{C}(O, 4)$.

2b) $\frac{z_C - z_A}{z_D - z_B} = \frac{-2 - 2\sqrt{3}i}{-2\sqrt{3} - 6i} = \frac{\sqrt{3}}{3}$, réel $\Rightarrow (AC) \parallel (BD)$.

2c) $AB = |-4 + 4i| = 4\sqrt{2}$ et $CD = |z_D - z_C| = 4\sqrt{2} \Rightarrow AB = CD$.

2d) Dans $ABDC$, $[AC] \parallel [BD]$ et $[AB], [DC]$ égaux $\Rightarrow$ trapèze isocèle.